CHAPTER 2
THE DESIGN PROCESS

2.1 DESIGN PROCESS CONTEXT

There are numerous variations of the design process, perhaps as many as there are designers. To try and place the following information into a common context, the design process structure used in *ASHRAE Guideline 0-2005, The Commissioning Process* (ASHRAE 2005a) will be used. For purposes of building commissioning, the acquisition of a building is assumed to flow through several broad phases: predesign, design, construction, and occupancy and operation. The design phase is often broken into conceptual design, schematic design, and design development subphases. Although the majority of design hours will be spent in the design development phase, each of these phases plays a critical role in a successful building project. Each phase should have input from the HVAC&R design team. The HVAC&R design team should strive to provide input during the earliest phases (when HVAC&R design input has historically been minimal) since these are the most critical to project success, as they set the stage for all subsequent work.

Design should start with a clear statement of *design intent*. In commissioning terms, the collective project intents form the Owner's Project Requirements (OPR) document. Intent is simply a declaration of the owner's (and design team's) needs and wants in terms of project outcomes. HVAC&R design intents might include exceptional energy efficiency, acceptable indoor air quality, low maintenance, high flexibility, and the like. Each design intent must be paired with a *design criterion*, which provides a benchmark for minimum acceptable performance relative to the intent. For example, an intent to provide thermal comfort might be benchmarked via a criterion that requires compliance with *ANSI/ASHRAE Standard 55-2004, Thermal Environmental Conditions for Human Occupancy* (ASHRAE 2004b), and an intent for energy efficiency might be benchmarked with a criterion that requires compliance with *ANSI/ASHRAE/IESNA Standard 90.1, Energy Standard for Buildings Except Low-Rise Residential Buildings*.

Design validation involves the use of a wide range of estimates, calculations, simulations, and related techniques to confirm that a chosen design option will in fact meet the appropriate design criteria. Design validation is essential to successful design; otherwise there is no connection between design intent and design decisions. *Pre- and post-occupancy validations* are also important to ensure that the construction process and ensuing operational procedures have delivered design intent. Such validations are a key aspect of building commissioning.

2.2 DESIGN VERSUS ANALYSIS

Anyone who has taken a course in mathematics or any of the physical sciences is familiar with the process of analysis. In a typical analysis, a set of parameters is given that completely describes a problem, and the solution (even if difficult to obtain) is unique. There is only one *correct* solution to the problem; all other answers are *wrong.*

Design problems are inherently different—much different. A design problem may or may not be completely defined (some of the parameters may be missing) and there are any number of potentially acceptable answers. Some solutions may be better than others, but there is no such thing as a single right answer to a design problem. There are degrees of quality to design problem solutions. Some solutions may be better (often in a qualitative or conceptual sense) than others from a particular viewpoint. For a different context or client, other solutions may be better. It is important to clearly understand the difference between *analysis* and *design.* If you are used to looking for *the* correct answer to a problem (via analysis), and are suddenly faced with problems that have several acceptable answers (via design), how do you decide which solution to select? Learn to use your judgment (or the advice of experienced colleagues) to weigh the merits of a number of solutions that seem to work for a particular design problem in order to select the best among them.

Figures 2-1 and 2-2 illustrate the analysis and design processes, respectively. Analysis proceeds in a generally unidirectional flow from given data to final answer with the aid of certain analytical tools. Design, however, is an iterative process. Although there are certain "givens" to start with, they are often not immutable but

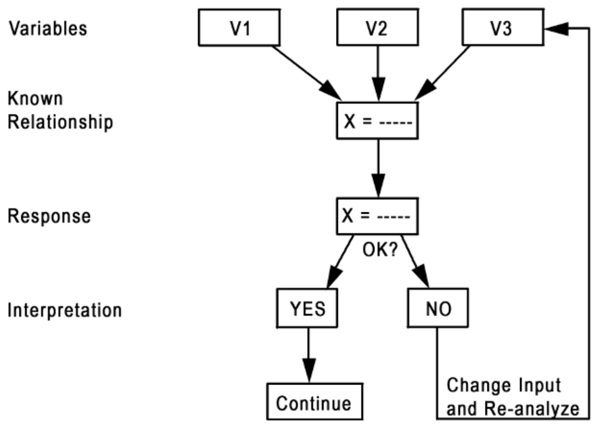

Figure 2-1. Diagram illustrating analysis.

subject to modification during the design process. For example, an owner or architect may be confronted with the energy implications of excessively large expanses of glass that had been originally specified and may decide to reduce the area of glazing or change the glazing properties. The mechanical designer may try various system components and control strategies before finding one that best suits the particular context and conditions. Thus, design consists of a continuous back-and-forth process as the designer selects from a universe of available systems, components, and control options to synthesize an optimum solution within the given constraints. This iterative design procedure incorporates analysis. Analysis is an important part of any design.

Since the first step in design is to map out the general boundaries within which solutions are to be found, it may be hard to know where and how to start because there is no background from which to make initial assumptions. To overcome this obstacle, make informed initial assumptions and improve on them through subsequent analysis. To assist you in making such initial assumptions, simple rules are given throughout the chapters in this manual, and illustrative examples are provided in the appendices.

2.3 DESIGN PHASES

A new engineer must understand how buildings are designed. Construction documents (working drawings and specifications)

2.3.1 Predesign Phase

Before a mechanical engineer can design an HVAC system, a building program must be created. This is usually prepared by the client or his/her consultants. The program establishes space needs and develops a project budget. The building program should include, but need not be limited to, the following:

- The client's objectives and strategies for the initial and future functional use of the building, whether it be a single- or multiple-family dwelling or a commercial, industrial, athletic, or other facility.
- A clear description of function(s) for each discrete area within the building.
- The number, distribution, and usage patterns of permanent occupants and visitors.
- The type, distribution, and usage patterns of owner-provided heat-producing equipment.
- The geographic site location, access means, and applicable building and zoning codes.
- The proposed building area, height, number of stories, and mechanized circulation requirements.
- The owner's capital cost and operating cost budgets.
- A clear statement of anticipated project schedule and/or time constraints.
- A clear statement of required or expected project quality.

Although some of this information may not be available before the mechanical designer starts to work, it must be obtained as soon as possible to ensure that only those HVAC&R systems that are compatible with the building program are considered.

While the architect prepares the general building program, the mechanical engineer has the responsibility of developing a discipline-specific program even though some of this information may be provided by the architect or owner. The building program and use profile provided by the owner or architect and the HVAC&R systems program developed by the engineer in response to the building functional program should be explicitly documented for future reference. This documentation is termed the *Owner's Project Requirements (OPR)* by ASHRAE Guideline 0-2005, and it provides the context for all design decisions. All changes made to the program during the design process should be recorded so that the documentation is always up-to-date.

The information that should be contained in the HVAC&R system program includes

- design outdoor dry-bulb and wet-bulb temperatures (absolute and coincident);
- heating and cooling degree-days/hours;
- design wind velocity (and direction) for winter and summer;
- applicable zoning, building, mechanical, fire, and energy codes; and
- rate structure, capacity, and characteristics of available utilities and fuels.

Additional information regarding solar radiation availability and subsurface conditions would be included if use of a solar thermal system or ground-source heat pump was anticipated.

The environmental conditions to be maintained for each building space should be defined by

- dry-bulb and wet-bulb temperatures during daytime occupied hours, nighttime occupied hours, and unoccupied hours;
- ventilation and indoor air quality requirements;
- any special conditions, such as heavy internal equipment loads, unusual lighting requirements, noise- and-vibration-free areas, humidity limits, and redundancy for life safety and security; and
- acceptable range of conditions for each of the above.

An understanding of the functional use for each area is essential to select appropriate HVAC&R systems and suitable control approaches because the capabilities of proposed systems must be evaluated and compared to the indoor environmental requirements. For example, if some rooms in a building require humidity control while others do not, the HVAC system must be able to provide humidification to areas requiring it without detriment to the building enclosure or other spaces. Some areas may require cooling, while others need only ventilation or heating. This will affect selection of an appropriate system.

2.3.2 Design Phase

In conventional (business-as-usual) building projects, serious work on HVAC&R system design typically occurs in the later stages of the design phase. Projects where energy efficiency and/or green building design are part of the intent or building types where HVAC&R systems are absolutely integral to building design (laboratories, hospitals, etc.), will see HVAC&R design begin earlier and play a more integrated role in design decision making.

The design phase is often broken down into three subphases: conceptual design, schematic design, and design development. The terms *schematic design*, *design development*, and *construction documents* are also commonly used to describe design process subphases. The purpose of conceptual/schematic design efforts is to develop an outline solution to the OPR that captures the owner's attention, gets his/her buy-in for further design efforts, and meets budget. Schematic (or early design development) design efforts should serve as proof of concept for the earliest design ideas as elements of the solution are further developed and locked into place. During later design development/construction documents, the final drawings and specifications are prepared as all design decisions are finalized and a complete analysis of system performance is undertaken.

The schematic/early design development stage should involve the preliminary selection and comparison of appropriate HVAC&R systems. All proposed systems must be able to maintain the environmental conditions for each space as defined in the OPR. The ability to provide adequate thermal zoning is a critical aspect of such capability. For each system considered during this phase, evaluate the relative space (and volume) requirements for equipment, ducts, and piping; fuel and/or electrical use and thermal storage requirements; initial and life-cycle costs; acoustical requirements and capabilities; compatibility with the building plan and the structural system; and the effects on indoor air quality, illumination, and aesthetics. Also consider energy code compliance and green design implications (as appropriate).

Early in the design phase, the HVAC&R designer may be asked to provide an evaluation of the impact of building envelope design options (vis-à-vis energy code compliance and trade-offs and/or green building intents), heavy lighting loads (i.e., more than 2 W/ft^2 [22 W/m^2]), and other unusual internal loads (i.e., more than 4 W/ft^2 [43 W/m^2]) on HVAC system performance and requirements. Questions should also be expected regarding the optimum location of major mechanical equipment—considering spatial efficiency, system effectiveness, aesthetics, and acoustical criteria. Depending upon the level of information available, the designer may be asked to prepare preliminary HVAC system sizing or performance estimates based upon patterns developed through experience or based upon results from similar, previously designed projects. Some design estiowner's prerogative. Design decisions should be clearly documented whether they constitute approval of, or deviation from, the designer's proposals.

The schematic HVAC design is used to coordinate critical HVAC&R system requirements with the architectural, structural, electrical, and fire protection systems, which are also in the schematic design stage, to resolve potential conflicts. Close integration of the mechanical and electrical systems with structure, plan, and building configuration requires the cooperation of all team members—architect, mechanical engineer, electrical engineer, structural engineer, acoustical consultant, and professionals from other disciplines. Such coordination and cooperation will extend to the other design phases.

The requirements of state or local building or energy codes (see Section 2.7) must also be considered at this point because of restrictions on the amount of glass, other envelope assembly requirements, lighting, HVAC&R system and equipment limitations, and, in some instances, on the annual building energy budget. If required, energy budgets are established. Some states or jurisdictions require a simplified prescriptive compliance calculation that can be prepared by hand; when annual energy-budget compliance calculations are required, however, these must be prepared by computer simulation. It is wise to select a computer program that will minimize the inputs required for both the load and the energy analyses—while providing acceptable accuracy. Programs are available that share input between loads and energy programs.

Architectural floor plans and elevations are developed in greater detail; structural, mechanical, and electrical systems are designed in compliance with applicable building codes; and drawings in preliminary form are prepared. Heat loss, heat gain, and ventilation calculations are refined and used to design the air and water distribution systems and to select equipment. System sizes and capacities are selected to match design and part-load conditions. The designer has the choice of sizing pipes and ducts manually or using a variety of computer programs. The main objective, however, is to develop system layouts for space requirements and cost estimates. If required by code or owner intent, more detailed energy studies are undertaken at this time and the accompanying calculations are performed. Construction details and cost estimates are refined based upon additional information. After preliminary drawings, outline specifications, and cost estimates have been approved by the owner, the project scope and solution are essen-